

YUHANG JIANG

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[Homepage](#) | [Google Scholar](#) | [GitHub](#)

Summary

- Double-degree MSc in Computer Science (University of Trento) and Data Science (SUPSI), EIT Manufacturing programme; Research Intern at Huawei Pisa Research Center working on multimodal agents and LLM post-training.
- Two first-author journal papers accepted in 2026: IEEE Transactions on Geoscience and Remote Sensing and TMLR.
- GeoSelect sets the training-free state of the art on RRSIS-D and RISBench; the Huawei multimodal agent memory work sets the state of the art on DMV-Bench and leads published memory-augmented agents on MemLens.
- Research focus: multimodal agents, LLM post-training, embodied AI and mechanistic interpretability, with several projects run independently from problem definition to paper.

Education

EIT Manufacturing Master & Doctoral School (Double Degree)

Data Science and AI for a Competitive Manufacturing · Scholarship: full tuition waiver + 10,000 EUR/yr

- **Università di Trento (UNITN), Italy** Sep 2023 - Mar 2026
MSc, Computer Science · GPA 98/110 · Degree awarded Mar 2026
- **University of Applied Sciences and Arts of Southern Switzerland (SUPSI)** Sep 2023 - Dec 2026
Master of Engineering (MSE), Data Science · GPA 5.1/6.0 · Thesis defended Mar 2026, diploma expected Dec 2026

Anhui University (Project 211), China

Sep 2019 - Jun 2023

BEng, Intelligent Science and Technology · Average 84.34/100

Scholarships: Academic Science and Technology Scholarship, First Prize (2021); Outstanding Academic Performance, Second Prize (2022)

Research Experience

Huawei Pisa Research Center

Apr 2026 - Oct 2026

Research Intern

Pisa, Italy

- **Led an independent research project on memory for multimodal agents**, from problem definition and experiment design to paper; first-author paper under review. Controlled experiments that hold the retrieved set fixed and change one variable at a time located the bottleneck in **delivery** rather than retrieval: delivering the original images is worth +13.87 points, perfect retrieval only +2.31, and the delivery gain grows on a stronger backbone (+17.92 at 32B).
- **Proposed DeliverMem, a training-free delivery method that leaves the memory store unchanged** (original modality, readable identifiers and temporal position, plus a retrieval-side adaptation): **state of the art on DMV-Bench**, ahead of the best prior method at every setting on both backbones by 7.63 to 22.62 points; on MemLens it leads the strongest published memory-augmented agent by at least 8.20 points at all four context lengths, on a tenth to a seventieth of the full-context input. At MemLens 32K with dev-fitted selection it **surpasses GPT-5.4 and matches Gemini-3.1-Pro** while delivering about 2,400 tokens, a tenth of what they read.
- **Owned the LLM post-training pipeline for cockpit AI end to end**: data construction and cleaning, SFT, RL with reward design, evaluation and result analysis.
- Built inference, training and evaluation pipelines on vLLM, Transformers and ms-swift, supporting large-scale controlled comparisons and ablations across backbones and context lengths.

Fondazione Bruno Kessler (FBK)

Mar 2025 - Mar 2026

Research Intern

Trentino-Alto Adige, Italy

- Proposed Active Instance Verification, a new embodied task, and built **PInVerify**, the first offline multi-view benchmark for it in photorealistic 3D environments.
- Built an MLLM agent with attribute decomposition, confidence-weighted multi-view tracking and next-best-view planning; fine-tuned via LoRA with SFT followed by RL alignment, reaching 85.6% accuracy.

Chengdu Jiaoda Guangmang Technology Co., Ltd.

Jul 2022 - Sep 2022

Algorithm Intern

Chengdu, China

- Industrial anomaly detection: GAN-based image reconstruction and object detection for defect inspection on high-speed rail components.

Publications

"GeoSelect: Spatial-Program Execution for Training-Free Referring Remote Sensing Image Segmentation"
IEEE Transactions on Geoscience and Remote Sensing (TGRS), 2026 · First author · [Project Page](#) · [arXiv:2607.03869](#)

Referring segmentation as typed spatial-program execution, no referring supervision: 58.86 / 55.27 mIoU on RRSIS-D / RISBench, over $2\times$ / about $1.7\times$ the best prior training-free results.

Yuhang Jiang, Guohui Deng*, Miaozhong Xu, Chao Ruan, Jinling Zhao, Linsheng Huang

"The Scissors Effect: When Resize-Based Input Diversity Helps or Hurts Transfer Attacks"

Transactions on Machine Learning Research (TMLR), 2026 · First author · [Project Page](#) · [arXiv:2606.22516](#) · [OpenReview](#)

Resize-based input diversity, a default in transfer attacks, reverses sign on adversarially trained surrogates, so using it blindly overstates robustness.

Yuhang Jiang, Xiaojing Chen

"PlnVerify: An Offline Embodied Benchmark for Active Instance Verification"

Poster, FMEA Workshop @ CVPR 2026 · Sole author · [Project Page](#) · [arXiv:2605.30639](#) · [OpenReview](#)

First benchmark for Active Instance Verification; active viewpoint selection gives no reliable gain, and detection quality binds the task.

Yuhang Jiang · Master's thesis project

Preprints and Other Publications

"A Circuit, Not The Circuit: Non-Unique Causal Localisation of the Mamba-2 State Sink"

[arXiv:2606.00930](#)

Names the state sink, the state-space analogue of the attention sink; a near-disjoint set of heads matches the causal effect of the probe-selected ones, so a probe recovers a circuit rather than the circuit.

Yuhang Jiang, Bowen Zhang

"Task Structure Reverses Layerwise State Encoding in Sequence Models"

[arXiv:2606.00926](#)

A layerwise probe profile reflects what the model was trained to predict, not its architecture; near-zero ablation drops do not show that a model ignores a feature.

Yuhang Jiang

"CL-RAG: A Closed-Loop Multimodal Retrieval-Augmented Generation Architecture for Robust Human-Robot Control Interaction"

WRC SARA 2025 (oral) · Bowen Zhang, Yuhang Jiang, Lingxiang Hu, Dun Li, Qianqian Hu*

"Improved lightweight identification of agricultural diseases based on MobileNetV3"

CAIBDA 2022 (oral) · Yuhang Jiang*, Wenping Tong

Note: * indicates the corresponding author.

Awards and Competitions

- **Kaggle Competitions Expert** (all solo): Silver, March Machine Learning Mania 2025 (64 / 1,727); Bronze, BirdCLEF+ 2025 (199 / 2,025); Bronze, BirdCLEF 2024 (70 / 974).
- **Mathematical Contest in Modeling (MCM), Finalist** (top 1%), 2021. Team leader; responsible for modelling and implementation. Youngest winner in the history of Anhui University.

Skills

- **Research areas:** multimodal LLMs, agent memory, LLM agents, RAG, embodied AI, mechanistic interpretability.
- **LLM post-training:** SFT, DPO, GRPO / GSPO, RL reward design, LoRA / PEFT, data construction and evaluation.
- **Frameworks and systems:** PyTorch, Transformers, vLLM, ms-swift; scikit-learn, MPI / OpenMP (HPC).
- **Programming:** Python, C/C++, Matlab, R.

Additional

- **Academic service:** Reviewer, ICLR 2027; InterpScience and TTCL Workshops @ NeurIPS 2026; FMEA Workshop @ CVPR 2026.
- **Languages:** English (fluent, TOEFL 102); Chinese (native).